

Instructions:

- Submit your answers as a single file (**.ZIP**) on or before the deadline provided in the LMS.
 - Submission must include this document explaining your code, UI and lessons learnt
 - Late submission will not be considered for the marking.
 - Make sure to include this document in your submission
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Student index: 22ug2-0035

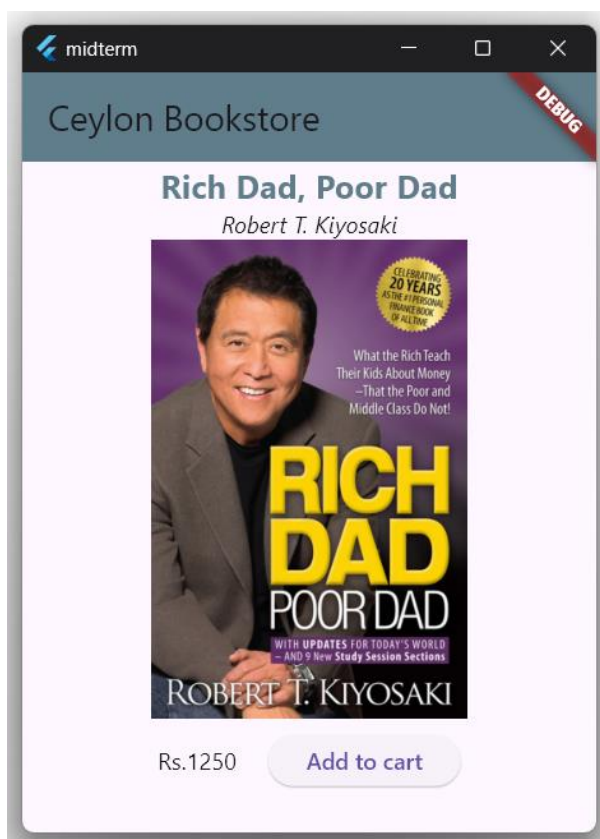
Date of submission: 29/01/2025

Understanding layout

The core of Flutter's layout mechanism is widgets. In Flutter, almost everything is a widget—even layout models are widgets. The images, icons, and text that you see in a Flutter app are all widgets. But things you don't see are also widgets, such as the rows, columns, and grids that arrange, constrain, and align the visible widgets. You create a layout by composing widgets to build more complex widgets.

Here's an example of a book store app, showing a list of books. Observe the completed UI (browser and emulator) and understand the breakdown. Create the Book layout as a custom widget, which can be reused.

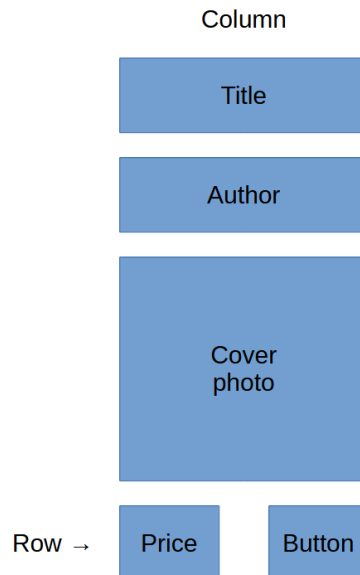
Single book



Browser

Widget layout

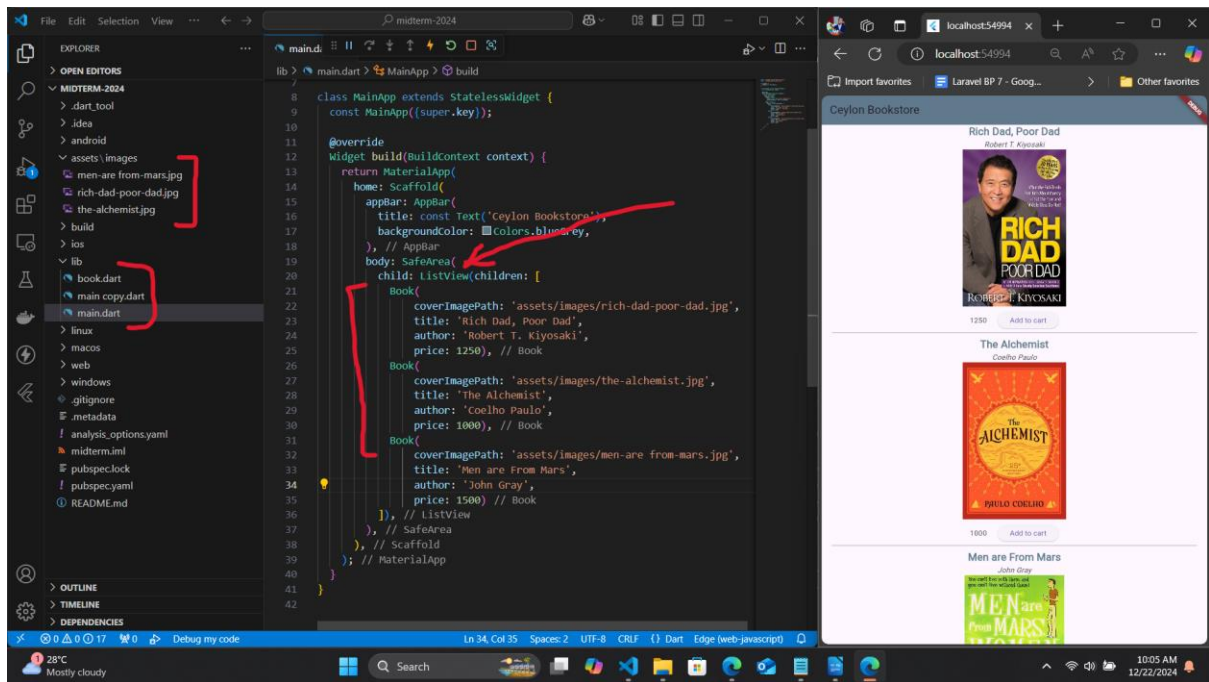
You may or may not use a Card to enclose the elements. The layout uses 4 elements inside a Column and 2 elements in the last Row. Text widget is used for Title, Author and Price.



Exercise: Basic Flutter UI (total marks: 100)

1. Draw the widget hierarchy diagram (20 marks)
2. Create the widget as per the UIs given (20 marks)
3. Create 3 child elements (objects) from the widget you created (20 marks)
4. Copy paste the code to your document (10 marks) or include a repo link
5. Copy paste the UI to your document (10 marks)
6. Write a note on lessons learned and your rationale for selecting components from the library (20 marks)
7. ZIP the whole project folder along with this document and upload to LMS

Sample code with a list of 3 books



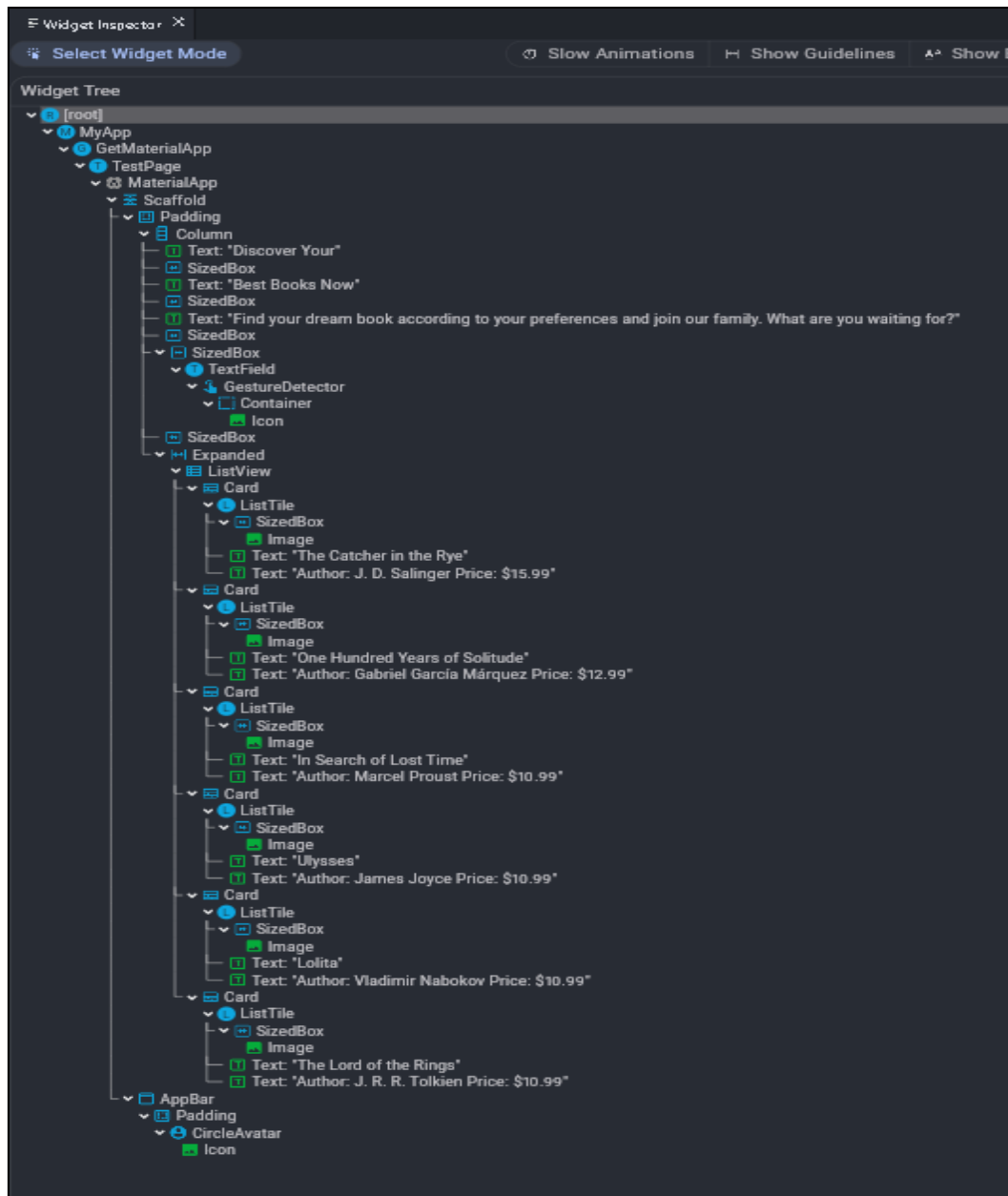
Read the documentation here:

- <https://docs.flutter.dev/ui/layout>
- <https://docs.flutter.dev/ui/layout/tutorial>
- <https://medium.com/flutter-community/flutter-layout-cheat-sheet-5363348d037e>
- <https://www.flutterbeads.com/set-background-image-in-flutter/>
- <https://www.flutterbeads.com/flutter-position-widget-in-stack/>

Paste the diagram & repo link here

You may post the architecture diagram here with a link to GitHub repo.

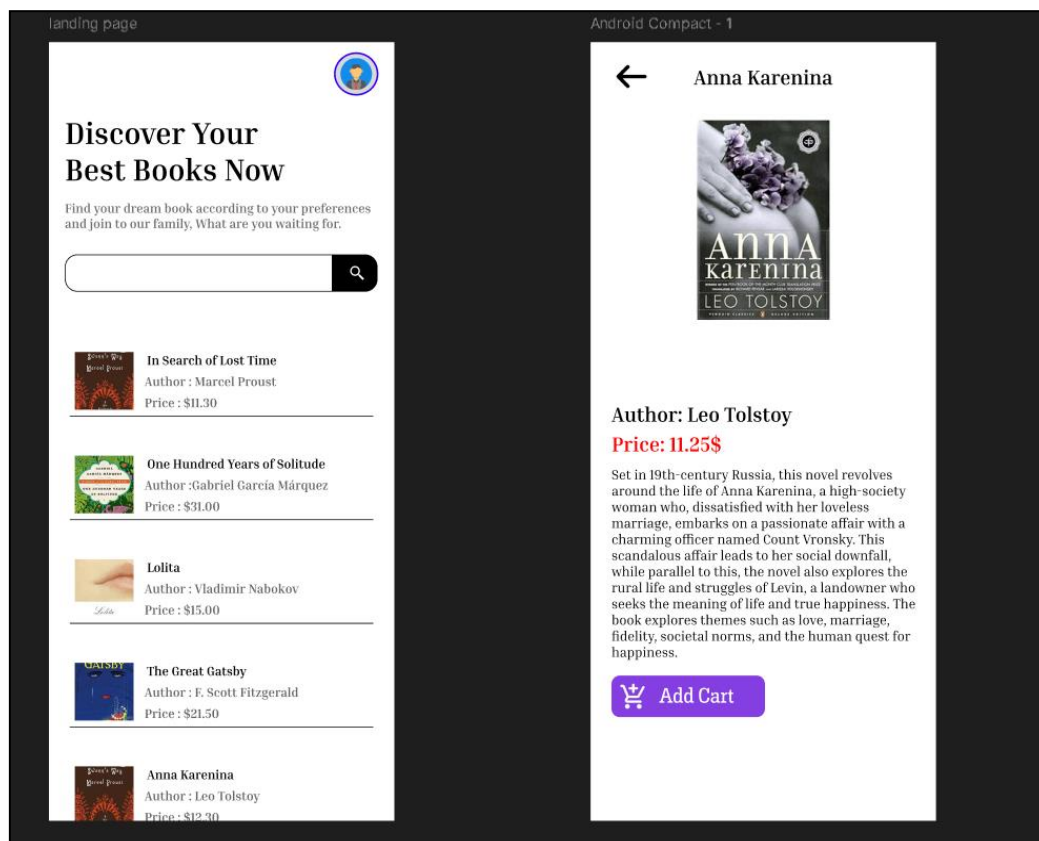
GitHub : <https://github.com/sahanHansaja026/Mid-term-project->



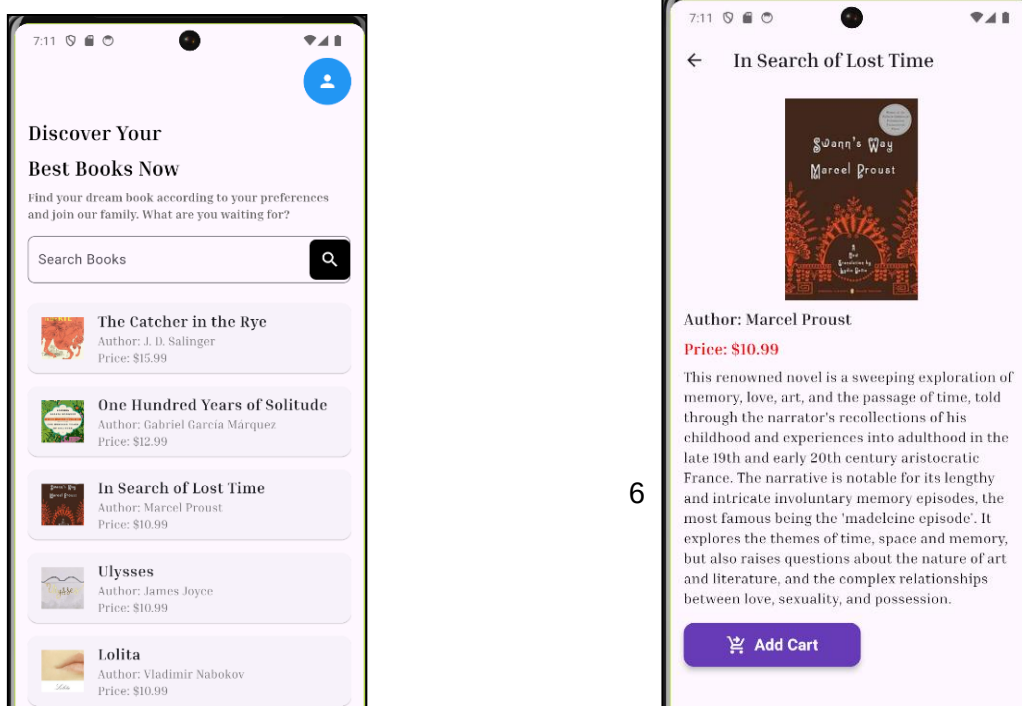
Paste the UI here

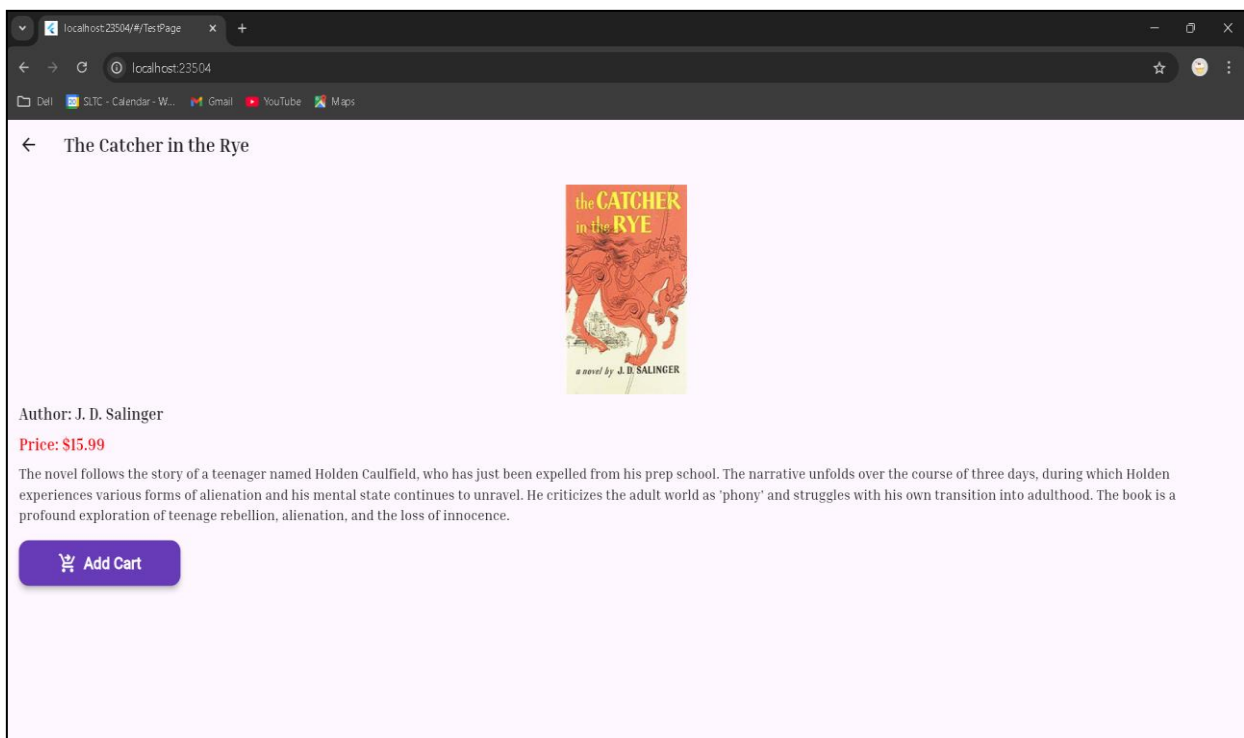
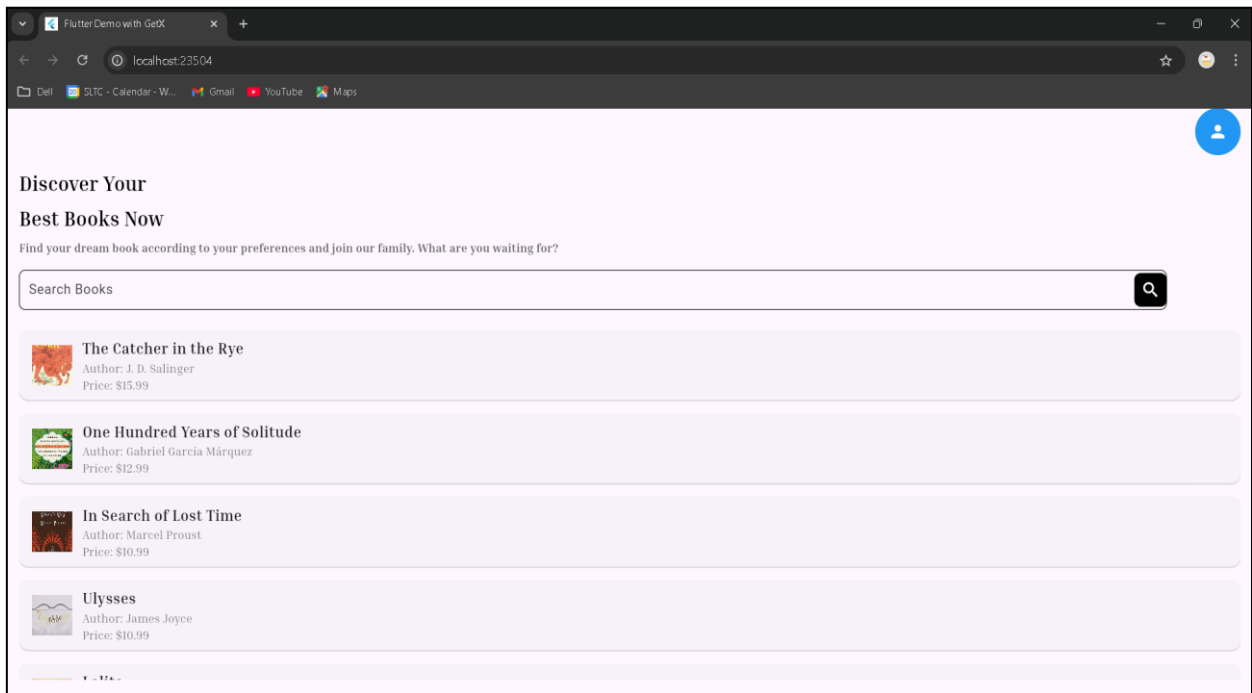
a. Figma prototype

Figma Link : <https://www.figma.com/design/uDa45jRVAt9ea1xoJJJJi0/Untitled>



b. Final Flutter UIs





Lessons learnt

Notes on what you learnt in doing this activity.

- **Widget Hierarchy:** Learn to visualize and structure Flutter widgets, arranging UI elements efficiently.
- **Custom Widgets:** Create reusable widgets to enhance modularity and maintainability (e.g., BookWidget).
- **OOP in Flutter:** Use custom widgets dynamically to populate UI with multiple instances.
- **Practical UI Building:** Implement functional UIs with widgets like Text, Column, Row, Card, and IconButton.
- **Coding & Debugging:** Write, test, and debug Flutter code across platforms, improving coding skills.
- **Version Control:** Document code, manage repositories, and follow best practices using GitHub.
- **UI Analysis:** Capture and evaluate app UIs to refine design and enhance user experience.
- **Reflect & Improve:** Reflect on lessons learned to justify design choices and improve future projects.